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(51) International classification :E04H0004120000, B63B0035320000, G01S0013340000, G01S0013931000, G01S0007350000 (86) International Application No :NA Filing Date :NA (87) International Publication No : NA (61) Patent of Addition to Application Number :NA Filing Date :NA (62) Divisional to Application Number :NA Filing Date :NA		(71)Name of Applicant : 1)Vardhaman College of Engineering Address of Applicant :Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- Name of Applicant : NA Address of Applicant : NA (72)Name of Inventor : 1)B.Pushpanjali Address of Applicant :Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- 2)B.Sai Ritheesh Address of Applicant :Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- 3)B.Alekhy Address of Applicant :Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- 4)B.Pranav Kumar Address of Applicant :Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- 5)Rajitha Ala Address of Applicant :Assistant Professor, Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- 6)A.Mounika Yesaswini Address of Applicant :Assistant Professor, Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- 7)Baini Anusha Rani Address of Applicant :Assistant Professor, Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad ----- 8)G Krishna Teja Address of Applicant :Assistant Professor, Vardhaman College of Engineering, Nagarguda - Shamshabad Road, Kacharam, Shamshabad - 501218, Hyderabad, Telangana, India. Hyderabad -----
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(57) Abstract :

An automated flood control and rescue device for water canals, comprises of a vertically extendable panel 101 with multiple extendable segments adjusts height based on real-time water level data from a radar sensor mounted on the panel 101, a motorized drawer arrangement 102, is controlled by an inbuilt microcontroller, actuates the panel 101 for optimized flood response, an extendable curved deflector flap 103 at the panel's 101 top edge deploys to redirect overflow, an AI-powered surveillance imaging unit 104 performs real-time object detection, a motorized horizontal slider 105 moves along the panel 101, carrying a rescue assembly 106 with an L-shaped hydraulic limb 107 and a cuboidal open-ended housing 109 equipped with a deployable lid 111, expandable base, and motorized pump 112, a L-shaped hydraulic rod 113 with a circular ring 115 and flexible net 117 collects floating debris via microcontroller-controlled sweeping and rotation.

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